An Agent-Based Study Assistant System

with OCR, Retrieval-Augmented Generation, and Text-to-Speech Tools

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Problem Context

Study is repeated transformation of fragmented course material.

PDFs / scans → Prepared text → Searchable sources → Study artefacts → Audio revision

- Students revise from lecture slides, scans, textbook extracts, notes, and copied fragments
- The same source must become notes, explanations, flashcards, quizzes, essay practice, or audio
- General chat alone hides provenance and makes long material hard to control
- Useful AI study support must remain source-grounded, modular, and inspectable

Core thesis

AI assistance is most useful when it coordinates the study workflow around the student's own material, rather than treating generation as a single unstructured prompt.

Problem Statement

1. Input readiness

Scanned slides and PDFs may be readable to a student but unusable by language tools until OCR produces text.

2. Grounded access

Long or fragmented notes require retrieval from prepared sources instead of uncontrolled prompting.

3. Output form

Revision requires different artefacts: explanation, active recall, essay practice, and listening.

Design challenge

Build a study assistant that moves from raw course material to source-grounded, reusable study artefacts.

Aim, Objectives, and Scope

Aim and objectives

- Coordinate OCR, retrieval, generation, and speech around realistic revision workflows
- Preserve source grounding across transformations
- Expose OCR, RAG, and TTS as standalone, inspectable tools
- Evaluate reliability, throughput, model suitability, and workflow feasibility

Claim boundary

- System-design and engineering thesis
- No claim of measured grade improvement
- No replacement for instructor feedback or student judgement
- Remote model quality, latency, cost, and privacy remain practical constraints

Study Workflow Requirements

OCR

Converts visual material into text. For study, quality includes order, equations, tables, and omissions.

RAG

Stores and retrieves the student's prepared material so answers remain tied to course-specific sources.

TTS

Turns prepared text into speech, after rewriting visual notation into a natural listening form.

PDFOCR
ordered JSONL page records

CHUNKVEC
marked chunks in SQLite vector store

CHUNKTTS
complete ordered `.opus` artefact

Each capability leaves an inspectable artefact rather than disappearing into a chat transcript.

Architecture: Flexible Agent, Deterministic Tools

INPUT
Student request + material

->

ORCHESTRATION
study-assistant selects one mode

->

OUTPUT
Study artefact or audio

INPUT READINESS
ocr-tool -> pdfocr -> ordered JSONL

GROUNDED ACCESS
rag-tool -> chunkvec -> SQLite vectors

OUTPUT MODALITY
tts-tool -> chunktts -> final audio

Agent role

Interpret study intent and select a mode such as transcription, notes, flashcards, quiz, essay, or audio.

Tool role

Enforce schemas, ordering, retries, exit codes, and artefact publication.

Engineering Decisions

Agent orchestration, deterministic tools

The agent owns study intent; tools own processing contracts, retries, schemas, and publication.

Ordered output over maximum raw speed

Concurrent requests may complete out of order, but published pages and chunks preserve source order.

Standalone command-line stages

OCR, retrieval, and speech can be run, inspected, cached, redirected, or tested independently.

Artefact-specific failure semantics

Partial OCR can be auditable; partial final audio is withheld because it may appear complete while omitting content.

Shared infrastructure

Nim tools share `relay`, `jsonx`, and `openai` for HTTP execution, JSON handling, and provider schemas.

Evaluation consequence

Behaviour can be checked through process channels, schemas, exit codes, stored artefacts, and ordered outputs.

Workflow Evaluation Method

PDF → **OCR text** → **Study mode output** → **RAG search** → **Speech artefact**

Evaluation criteria

- Source grounding against prepared material
- Appropriateness to the selected study mode
- Inspectable intermediate artefacts
- Operational reproducibility of the run
- Visible limitations rather than hidden failures

Recorded workflows

Essay practice

Association Analysis slides -> exam prompts and sample answers

RAG revision

Textbook section -> stored chunks -> targeted Naive Bayes

Flashcards

Anomaly Detection slides -> active-recall cards

Notes to audio

Clustering notes -> speech-ready text -> final audio

Testing Strategy and Evaluation Credibility

Deterministic evidence

- `pdfocr`: page selection, request IDs, retry queue, JSON result schema
- `chunkvec`: chunk parser, embeddings configuration, SQLite/vector integration
- `chunktts`: chunk splitting, retry pressure, audio validation, finalisation
- `relay`, `jsonx`, `openai`: transport, parsing, and provider-schema helpers

Evaluation principle

- Separate local implementation correctness from live provider variability
- Use live model runs as empirical operational evidence
- Verify that local tools still preserve ordering, failure semantics, and artefact boundaries
- Treat limitations as part of the evidence, not as an afterthought

Evaluation: OCR Throughput

72-PAGE SLIDE PDF BENCHMARK

Bounded concurrency made recorded OCR throughput practical while preserving ordered output.

15.89x

speedup

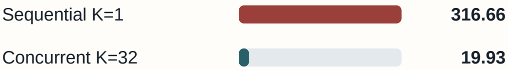
K=32 vs sequential K=1

93.71%

relative time reduction

296.73 seconds saved

RUNTIME COMPARISON



Config	Runtime	Result
K=1	316.66 s	72/72 ok
K=32	19.93 s	72/72 ok

Operational measurement; depends on provider latency, rate limits, network conditions, and document complexity.

Evaluation: OCR Model Benchmark

68 ACADEMIC PAGES FROM 34 PDFS

Model	CER	WER	Order F1	Math F1
PaddleOCR-VL	0.4634	0.4732	0.3751	0.7248
olmOCR 2	0.4682	0.4893	0.3252	0.6743
DeepSeek-OCR	0.5862	0.6512	0.1452	0.4684

Model	Total cost	Cost/page
PaddleOCR-VL	$0.0420	$0.0006180
olmOCR 2	$0.0214	$0.0003142
DeepSeek-OCR	$0.0041	$0.0000597

Benchmark character

Human gold labels; equations, tables, diagrams, and multi-column academic layouts.

Selection tradeoff

PaddleOCR-VL leads strict accuracy. DeepSeek-OCR is cheapest. olmOCR 2 is selected for recall-oriented suitability at lower cost than the strict-accuracy winner.

Workflow Evaluation Results

Essay practice

Association Analysis slides produced four exam-style prompts with sample answers covering Apriori, support, confidence, lift, and interestingness.

Flashcards

Anomaly Detection slides produced 25 active-recall cards covering definitions, settings, methods, limitations, and applications.

RAG exam revision

Textbook pages 358-402 were OCR-processed and stored as 26 semantic chunks; search retrieved Naive Bayes zero-probability handling.

Study notes to audio

Clustering notes became study notes, speech-ready text, and a final `.opus` file from 24 ordered speech chunks.

CRITICAL OBSERVATIONS

RAG should preserve retrieved chunks alongside synthesis; formula and diagram uncertainty should be flagged; flashcards and notes need coverage, duplication, and ambiguity checks.

Critical Interpretation and Threats to Validity

Interpretation

- Modularity makes AI study workflows easier to inspect and test
- Artefact boundaries create practical reliability checkpoints
- Bounded concurrency improves network-bound OCR in the recorded benchmark
- Retrieval quality depends on disciplined chunking, metadata, and evidence capture

Threats and limitations

- No controlled learning-outcome or retention study
- RAG evaluation does not yet use labelled query sets
- TTS evaluation verifies artefact correctness, not subjective naturalness
- OCR and throughput results are dataset- and provider-dependent
- Model behaviour, pricing, latency, and availability can drift

Contributions

Conceptual

Frames study assistance as a source-grounded workflow from raw material to revision artefacts.

Technical

Implements OCR, semantic retrieval, and speech pipelines with ordering, retries, schemas, and exit contracts.

Architectural

Separates agent orchestration, tool definitions, Nim executables, and shared libraries.

Evaluation

Combines contract tests, throughput evidence, OCR model comparison, and recorded workflow critique.

Future Work

Near-term engineering

- Run manifests linking source PDFs, OCR cache entries, retrieved chunks, generated text, TTS input, and final audio
- Labelled retrieval evaluation and stored retrieved evidence
- Broader OCR corpora and repeated measurements
- Validation passes for formulae, diagrams, duplicate cards, and coverage gaps

Broader validation

- Provider/model selection policies for cost, latency, quality, and privacy
- Local or private backends for sensitive course material
- Human evaluation of artefact usefulness
- Learning-outcome studies for retention and exam preparation

Conclusion

Main conclusion

`study-assistant` shows that AI study support can be practical and academically inspectable when flexible generation is constrained by source grounding, modular tools, and explicit artefact contracts.

- OCR improves input readiness
- RAG supports source-grounded access
- Study modes and TTS support different revision forms
- The architecture makes processing visible and testable

72/72
OCR pages succeeded
throughput benchmark

26
semantic RAG chunks
textbook revision workflow

24
speech chunks
notes-to-audio workflow

Questions

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